

Report on Logistic Support

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Introduction

The Army team was responsible for providing Logistic Support to the XII Expedition and X winter-over team from December 1992 to February 1994. The logistic support included the operation, maintenance and repair of the life-support systems, vehicles, generators, running of service convoys, handling of fuel, support in the conduction of scientific activities, medical cover and other miscellaneous tasks as allotted by the station commander and/or DOD.

The army team comprised of 16 members – one officer and seven ORs from Engrs, one officer and six ORs from EME and one officer from AMC. The report on the Medical cover has been submitted separately by the AMC officer.

Summer Period

Tasks

Besides the various tasks inherent to such an expedition, the XII Expedition had only one major summer task of constructing Genset accommodation and installing four 62.5 kVA generators to overcome the critical situation arising out of the fire accident of the IX Wintering team wherein freshly prepared power house was totally gutted. The preparation with R&DE (Engrs) during the summer period has been narrated in this report.

The construction of three containers, each 9 m × 3 m × 3 m – two for housing two generators each and one for supervision and monitoring purpose were undertaken. The various jobs involved were :

- (i) Selection of site
- (ii) Levelling of the area
- (iii) Foundation work

- (iv) Assembling of basic structure
- (v) Insulation and fireproofing
- (vi) Construction of hexagonal connection block
- (vii) Installation of accessories
- (viii) Domestic wiring.

The site selected and prepared by the previous team was approximately 400 m from the main station and considered too far. A fresh site about 100 m from the station was selected and prepared. It is interesting to note that practically all areas around Maitri can be levelled easily and accurately with minimal dozer effort.

The foundation work was rather complex and time consuming taking as much as ten days. The foundation was based on cribbiers resting on cement blocks. The use of hydraulic binders provided by R&DE (Engrs) was found to be very effective technique for such work. In fact, similar grouting technique with conventional cement was also tried out and found very convenient. The time spent in accurate foundation work was well rewarded in terms of quick and trouble free assembly of the knocked down containers.

Having practiced the construction at R&DE (Engrs), the assembly of containers posed no problem and the same was accomplished in bare five days. The crane proved to be very useful especially when the construction had to be carried out with limited manpower of five to six Engrs persons.

Snow accumulation around any structure seems to be a major problem in Antarctica. Hence, any structure constructed at ground level is likely to get buried during blizzards. To avoid such a problem there is a need to provide adequate ventilation under such structures. It would have been a good idea if these containers were planned to be constructed on stilt about 1 to 2 metres high. This may be noted for future.

The insulation and fire proofing is time consuming as a lot of improvisation and fine work had to be carried out at site. More than fifty percent of the construction time was taken up by this work. Again the construction of the connecting hexagonal structure and the domestic wiring took a lot of time. We need to develop a design by which all such are included within the prefabricated components.

An interse comparison of the three containers is given below:

The Walchand container was quickest to construct due to its true pre-fabricated construction, though apparently it is not as strong as the other two, it is estimated to be strong enough for the environment available at Maitri. However, the problem of sealing the joints has to be solved.

The Transfreight container is considered sturdiest and easy to construct but too heavy. It also has major sealing problems. A modified structure with all panels akin to the main doors may be a good idea to experiment with.

The Raj Power container is considered too heavy and too cumbersome to be categorised as a prefabricated structure.

Installation and Commissioning of Generators

The work involved the following jobs :

- (i) Placement of two gensets each in two containers
- (ii) Fixing of exhaust systems
- (iii) Provision of fuel supply
- (iv) Making electrical connection
- (v) Testing of gensets on load.

Unfortunately, the above work was not rehearsed at R&DE (Engrs) due to paucity of time. Hence each job was very time consuming. Had all the connections been pre-prepared and tested during the preparatory stage, valuable time would have been saved in Antarctica.

In spite of the meagre manpower and the problems encountered during the execution stage, the complete task was completed and the new power house 'Aditya' was commissioned on 18 February 1993. Certain finishing work such as domestic wiring, roof sealing, heating systems and hexagonal connecting block shall be taken up subsequently. The success of the task is owed to the small band of the Army team and the officer from R&DE (Engrs) who toiled many times late into the night to meet the deadline.

Repair and Recommissioning of the Cold Storage

Initially an attempt was made to repair the existing deep freezers but it had to be abandoned due to lack of manpower and paucity of time. As an alternate measure, the new deep freezers brought by this team were installed and both cold storages were made functional. Now both are capable of holding foodstuff at -20°C .

Taking Over of Vehicles

While taking over the fleet, essential and outstanding repairs were undertaken to ensure full readiness for the ensuing convoys. A number of vehicles required tyre change which was attended to. The two vehicles brought by our team developed severe electrical snag and took a lot of time and effort to repair. The need to carry out the overhaul at 512 bases workshop well in time and

returning the vehicles to Antarctica only after thorough testing cannot be over emphasised.

Taking Over of Life Support Systems

The various systems to be taken over were

- (i) Generators
- (ii) Electrical supply
- (iii) Central heating
- (iv) Water supply
- (v) Waste disposal

The generators and the electrical supply systems were taken over by the EME personnel well in time. Certain major repairs and maintenance tasks were carried out on the generators with the spares brought by this team. It shall be our endeavour to keep these generators in as healthy a state as possible so that they can act as a standby power house for the new one.

The central heating, water supply and waste disposal systems had to be taken over with just a cursory briefing as all the personnels were fully occupied in the priority construction tasks. An attempt was made to request the previous wintering team to carry out the briefing and on the job training from 5 to 20 Feb. 1993 when the construction activities were estimated to ease out. However, this help was not forthcoming from them since they had handed over the station and left for the ship on 1 Feb. 93. To obviate such an undesirable situation, either the summer tasks should be so planned that the wintering team has time to carry out the necessary on the job training on the life support systems or a spirit be cultivated in the previous team that their expedition is not over the day they hand over the station.

Receipt of Fuel

At the time of taking over the station on 1 Feb. 93, a meagre quantity of approximately 16 kl of fuel, adequate for about 12 days was handed over in location. Some 60 kl of fuel lying buried in snow and ice from 1 to 3 km from Maitri was neither visible nor considered retrievable by this team. Hence, heli-lifting of fuel gained urgency. Approximately 60 kl enough for two months were air lifted to Maitri. The remaining was unloaded at the shelf and at DG for collection by convoys in due course of time. For shifting fuel to DG the two vehicles brought with this team were extensively used for carrying out shuttle convoys. Retrieval of bulk fuel tanks from their buried state at DG and preparing them for decanting of fuel at the shelf was a major tasks.

Winter Period

The following work was undertaken during the winter period :

- (a) Completion of pending jobs in Aditya
- (b) Life-support systems
- (c) Construction and maintenance works
- (d) Maintenance of vehicle fleet and running of surface convoys
- (e) Fuel management
- (f) Support to scientific activities
- (g) Other miscellaneous tasks
- (h) Summary of recommendations

Aditya Power Complex

While the main structure of the knock down containers and the installation of the generators was completed and the power complex was made operational on the 18 Feb. 93, there remained a number of small jobs. These included sealing of walls and roofs, installation of fuel tanks and the fuel supply system, securing of power cables on poles, completion of earthing pits and construction of a connecting hexagonal block between the three containers were done subsequently. These jobs were completed only by mid April 93 as they were not only very time consuming, but also required a lot of manpower.

Having observed the structure over the winter period, the following observations and recommendations are made :-

- (a) Any structure in Antarctica must be founded on stilt type foundation as was done in the case of the main station. Structures founded directly on the ground or on box grillage create severe snow accumulation problems. Non-adherence to this principle in case of all structures at Maitri creates severe problems during blizzards.
- (b) Steel structures being very heavy are difficult to handle and work with. Sealing gaps between the steel panels pose a great problem. As long as adequate fire-proofing is done along the electrical wiring and fuel lines, prefabricated sandwiched insulated wooden panels are best for such structures. Alternatively, the structure should be based on pre-constructed ISO-container modules.
- (c) Problem of providing adequate ventilation for generators and at the same time preventing snow ingress during blizzards needs to be looked into.

- (d) Due to inherent heat produced by the running generators no additional heating is required for the generator complex.

Life Support Systems

Power Supply

Presently electrical power supply is being provided by two separate complexes: 'A' Block with four old 62.5 kVA generators and Aditya with four new 62.5 kVA generators. The 30 kVA generator located in the container is used as a back-up and for welding jobs. Other small generators are used for out-station activities. The generators in the 'A' Block are old and one of them had gone out of action. It had to be overhauled completely and required replacement of the crank-shaft. A crank-shaft was removed from an old generator at DG and brought out through a narrow hatch from 50 ft beneath the snow.

Heating System

The central heating system was kept fully operational by carrying out all the routine and scheduled maintenance. Considerable fuel economy was effected by maintaining an overall lower temperature in the station. The pleasant temperate temperature of 18 to 22°C was felt to be more comfortable and better from the health point of view. Experience has shown that the burning of boilers as per the schedule given below is more than adequate :-

- | | |
|-----------------------------------|---|
| (a) Nov to Feb | One boiler for 2 to 3 hrs daily for maintenance purpose only. |
| (b) Sept to Oct
and Mar to Apr | One boiler for 6 to 10 hrs daily. |
| (c) May to Aug | One boiler for 12 to 24 hrs daily. |

In times of blizzards or high wind conditions the above timings had to be modified to achieve the desired temperatures in the station. Care had to be taken to keep the circulation pump working to prevent air-locks and freezing of water in the pipes.

Following major works were carried out besides the routine maintenance in the heating system :

- (a) Change of heating coils in boiler nos. 1, 2 and 4
- (b) Fitting of hour totaliser for burners and circulation pumps in all the boilers
- (c) Flushing of whole system, including the repainting of the expansion tank twice

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- (d) Recasing and recapping of electrical wiring to reduce fire hazards.

It is felt that the present radiators are too bulky and not efficient in heat transfer. Also being of iron, they tend to corrode very fast, leading to leakages. Radiators used in DG station and Novo-lazarevskaya, the Russian station may be examined for an improved design. Also the iron expansion tank and duct heating tanks should be replaced by stainless steel tanks to prevent rusting and allied problems.

Water Supply System

The centrifugal pump was used as the mainstay for the water supply with the submersible pump as the standby. The submersible pump was operated once a fortnight to check and maintain its functioning. Some of the works carried out to improve the system were

- (a) Recapping and recasing of the electrical wiring with fire-proof bison board base to reduce fire hazard.
- (b) Installation of additional immersion rods in the heating tanks to maintain 30 to 40°C temperatures for hot water supply during peak demand hours in the mornings and evenings
- (c) Flushing and cleaning of hot and cold water storage tanks
- (d) Provision of a direct copper pipe line to the kitchen to ensure rust free water supply

Cold Storage and Refrigeration

In Maitri there are two walk-in cold stores and nine units of small household deep-freezers. On arrival of our team, both the walk-in cold storage units were made functional by installing new refrigeration units brought with us. By July, both these refrigeration units had become non-functional and the frozen food stuffs had to be shifted outside. One of the units were made functional by October with the refilling of the R-502 gas brought from DG. The foodstuffs that were being stored in the open during the winters could thus be saved before the impending summer. The second unit too was made fully operational before handing over the station on the 1 Feb. 94.

Waste Disposal

With the aim of abiding by the Antarctic norms regarding environment, both in letter and in spirit, conscientious efforts were made towards the

streamlining the waste disposal system at Maitri. Plastics, toilet-ash, metal, glass and left-over food wastes were duly segregated and sealed in barrels for back-loading to India. Paper waste was burnt from time to time and ash collected and sealed in empty barrels for back-loading.

The two Klargesters for processing the waste water were kept functional all the time. During Jan.-Feb. 93, a new waste water pond was made to prevent the effluent from reaching the lake. The outlet pipes of the Klargesters were changed with iron pipes to prevent leakage and to reduce the trace heating load. In Jan. 94, this pond has been further enlarged and the bund towards the lake widened and raised to prevent even the surface flow towards the lake.

Construction and Maintenance Works

Station Office

The station office was repainted and renovated. Racks for files, books and stationary, side boards for TV, VCR and refrigerator, large size station commander's table and computer table were fabricated and installed.

Main Corridor and Lounge

The wooden floor was redesigned and refixed to eliminate disturbing noise. All the brass expedition name plates and photographs were framed and refixed. The electric wires were recapped and recased to improve fire safety and appearance. The upholstery, curtains and carpets were changed. The lounge and the corridor were repainted.

Bathroom and Toilet Blocks

The bathing units were provided with showers hot and cold water mixing facilities. The urinals were changed and flooring was redone to ensure better cleanliness. All five toilet sheets and incinerators were changed. The exhausts were modified to prevent smoke leakage from incinerators. Fire proofing was also improved by fixing gypsum boards.

Rear Cold Porch

It was seen that cold draught was coming in from the rear door, which now have to be opened frequently due to the new power house having come up. To overcome this problem a double door airlock of 4 ft × 8 ft was made along the rear door.

Darkroom

The old darkroom was found to be unsuitable because of lack of heating and water supply and disposal problems. Hence one of the chemical toilet module located in the bathroom block was modified into a darkroom with all requisite facilities.

Storage Racks

There being total lack of storage facility for the pumps, motors and the toilet incinerators spares, storage racks were constructed inside and behind the chemical toilet module for this purpose.

Living Rooms

The living rooms were improved by providing side boards and writing tables. All the rooms either were painted or wall papered. Some rooms were also provided with pot holders for growing room plants.

Vehicle Maintenance and Surface Convoys

Vehicle maintenance was done regularly as per guidelines issued by the manufacturer, irrespective of weather conditions, polar day/night situations and non-availability of a garage. All the vehicles have been handed over to the new team in perfect working condition. In many cases spares have been retrieved from unserviceable assemblies or from DG hangar and used for repairs thus reducing maintenance costs and requirement of fresh spares.

A total of 13 convoys had been carried out during the period of wintering over. These included convoys for transportation of fuel and cargo, conveyance of empty containers from DG to Maitri, scientific activities at the shelf and backloading of unserviceable items and burnt out generator shed remnants. The first ever mid-winter convoy for biological sampling was done in June 93 from Maitri to the shelf and back in a record time of 23 hours and 50 minutes. The route from Maitri to the shelf has been extremely well marked by placing barrels at 3 minute driving intervals to ensure that even in severe blizzard conditions or during polar nights the drivers will not get lost. The route is safe and short and now even the Russians from Novo station have started following it upto DG. Three joint convoys have been conducted with the new team and the shortest and safest route to the shelf has been shown to them on ground – a distance which had been covered in 7 hours and 10 minutes by the X winter over team.

Fuel Management

Taking into account the 16 kl handed over by the previous team and the fuel brought by our team and the losses due to transportation and leakage, our team had approximately 330 kl of jet A-1 fuel at its disposal. Having handed over the fuel in excess 50 kl to the new team, we were able to manage our expedition within 280 kl. This was possible only by effecting strict economy measures such as reduced electrical works enabling shutting down of one generator from Oct to Dec and lower living temperatures in the station enabling reduced boiler burning hours as given above. Thus it is seen that fuel consumption from each requirement of 28 kl in winter can be cut down to 10 to 12 kl in summer months thus enabling a monthly average of 18 to 20 kl. If and when larger generators are inducted further fuel economy will be achieved.

It is felt that there is a requirement of larger bulk storage at Maitri and the shelf as in the case of neighbouring Russian station. This will reduce the need of hurried convoys and related work pressure during summer period. The practice of conveying and storing fuel in barrels must be discontinued not only are they difficult to carry and difficult to fuel out from; they also tend to get buried under snow during winter.

Support to Scientific Activities

Full support was given to the scientists of the X wintering team and also to the XIII expedition to enable them to successfully complete their tasks as per laid down schedules :

(a) NIO

- (i) Manufacture of grab and winching tripod for lake/sea sampling.
- (ii) Fitment of facilities for wet and dry samples analysis inside Maitri station.
- (iii) Conduct of monthly convoys to the shelf for sampling including during mid-winter months.
- (iv) Vehicular and other assistance in lake sampling as and when required.

(b) GSI

- (i) Vehicular, generator and manpower assistance in ice core drilling continuously for one month, wherein they achieved a record depth of 60 metres in winter months.
- (ii) Repair of imported ice core machine every now and then.
- (iii) Vehicular assistance in GSI stakes monitoring and field camps.

- (iv) Preparation and testing of Weddell Sea hut and Pneumatic tents for field camps.
- (c) IMD
 - (i) Fitment of balloon treatment room in erstwhile chemical toilet module.
 - (ii) Assistance in launching of balloon in high wind conditions.
- (d) R&DE(E)

Assistance in taking structural and other observations as requested by the R&DE(E) representative.
- (e) PRL
 - (i) Modification of one HIAL TECH hut with overhead perspex cubicle observation box.
 - (ii) Provision of side boards for fitting of instruments.
- (f) NPL

Construction of working table and fitment of teflon observation window for MM wave experiments.
- (g) NEERI
 - (i) Fitment of laboratory facilities.
 - (ii) Assistance in taking environmental observation both around Maitri and in lake.

Other Miscellaneous Works

Fire Fighting

The responsibility of fire precaution and fire fighting was allotted to members of Army Team. A detailed SOP was formulated laying down the precautions, alarm system, groups and drills for fire fighting. A strict vigilance was maintained specially in keeping the heat/smoke alarms and fire equipment functional at all times. Fire fighting practice was carried out regularly. The point of success in the task lies in the fact that there was no fire minor or major during the entire wintering period.

Area Cleaning

All members of the team devotedly worked towards cleaning up the area around Maitri. All unwanted rubbish and materials were collected and disposed

off suitably. Beside the weekly Saturday "Shramdaan" days, a few extended cleaning campaigns were carried along with other members of the team.

Recovery of Containers from DG

Three 20 kl bulk fuel tanks and five storage containers buried in blue ice since 1988 were retrieved in extremely trying weather conditions and mounted on sledges without the availability of a crane. The fuel tanks were transported to the shelf and the containers were taken to Maitri. This whole exercise has resulted in the saving of valuable foreign exchange.

Rescue Operation

A rescue convoy consisting of Lt Col DVS Rathee, Lt Col G Ravindranath, HMT K Sadanandan and HMT Vilas Shinde deputed by the Station Commander on request from Russian Expedition leader set out on a search and rescue mission on 25 May 1993. Due to extreme weather conditions two attempts failed. Finally on 27 May 1993 the team rescued the three stranded Russian expedition members at a distance of 20 km from Maitri and 15 km from their station in heavy blizzard and poor visibility conditions. They were provided with first aids and medical comforts.

Repositioning of Portacabins and Genset Accommodation at DG

Porta-cabin, the 30 kVA genset accommodation and Pistenbulley cabin at DG were found to be almost completely buried under the snow. As these items were being used by the scientists during the summer, they were dug out and repositioned on empty fuel barrels so that they remain usable for a few more seasons. Toilet module mounted on sledge also partially buried in snow, was also shifted to a suitable place.

Spares Management

Vehicle and generator spares lying at different places in and around Maitri, hangar and DG were collected and stored at one place. They were properly catalogued and hence it was possible to project a realistic demand for fresh spares resulting in considerable reduction in Government expenditure.

Manpower Assistance to XIII Expedition

All possible assistance in store handling, repairs and maintenance were provided to the incoming team both at Maitri and on board ship.

Recommendations

1. A separate accommodation should be constructed for lodging spares, assemblies and construction materials.
2. It is strongly felt that to ensure optimum utilisation of power, larger capacity generators of 250 kVA should be inducted and smaller capacity generators of 62.5 kVA inducted every year should be done away with as they require greater space, manning effort, more manpower, maintenance and fuel cost. One 250 kVA generator shall cater for all the power requirement of winter, summer and for future expansion.
3. Induction of different types of vehicles should be discontinued as it causes problems of maintenance, spares inventory, SMTs and tools and technical knowhow.
4. Greater cognisance should be given to the demands prepared by the winter over team as they carry out a major exercise to prepare a realistic and practical demand list. It is seen that spares and equipment being sent are both much more and other than those demanded.
5. The vehicles and other equipment backloaded should be overhauled promptly and returned with the next team.
6. As surface transport is going to be major mode of carriage of stores, there is a need for giving monetary incentive to the personnel manning the convoys to offset the difficult conditions as being done by other stations.
7. Suitable inclusion needs to be made in the Deputation order to enable service personnel to draw ration allowance in lieu of their entitled rations as applicable.
8. Antarctic expedition members should be exempted from income tax for the period spent away from India as being done for other postings/deputations abroad.